Roll No. ....

TME–501(A)

B. TECH. (FIFTH SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

HEAT TRANSFER

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Classify heat exchangers, and draw a neat sketch for each. (CO4)

(b) A cylinder of diameter 5 cm and length 25 cm is insulated on the outer surface so that heat transfer takes place in axial direction only. If the temperature of the two ends of the cylinder is maintained at 1350°C and 50°C and the thermal conductivity of the metal varies as $k = 0.838$ $(1 + 0.0007T)$ where Tin °C. Calculate the heat transfer rate per square meter through the cylinder. (CO3)

(c) Define the terms absorvity, reflectivity, transmissivity, black body and and real body. (CO6)

2. (a) Derive the expression for rectangular fin in case of heat dissipation from FIN Ainsulated at the tip. (CO2)

(b) A 5 cm diameter and 15 cm long rod (k = 20W/mK) protrudes out from a heat source at 100°C into an ambient at 25°C with a convection coefficient of 8 W/m² K. Determine : (CO3)

(i) Temperature distribution in the rod

(ii) Temperature at the free end

(iii) Heat flow out the source

(iv) Heat flow rate at the free end

(c) How calculation of surface area is done in case of heat transfer and explain the concept of fouling. (CO4)

3. (a) What do you mean by free and forced convection explain in detail with boundary layer development phenomenon over a flat plate. (CO3)

(b) Two concentric spheres of diameter 20 cm and 30 cm separated by air at one atmospheric pressure. The surface temperatures of the two spheres are 320 K and 280 K respectively. Determine the heat transfer by natural convection from the inner sphere to the outer sphere. Use the relation

$Nu = 0.228Ra^{0.226ssss}$

Properties at mean film temperature are

$K = 0.026W/m\text{-}K$

Kinematic viscosity = $1.57 * 10^{-5} * m^2/s$

$Pr = 0.712$

(c) Explain the following dimensionless numbers with their physical significance (i) Prandtl number (ii) Reynolds' number (iii) Grashoff number (iv) Nusselt number.

4. (a) Derive the expression for LMTD for parallel flow heat exchanger.

(CO5)

(b) The both ends of a 6 mm diameter copper rod (U-shaped) having thermal conductivity = 330 W/mK are rigidly connected to a vertical wall as the wall temperature is constant at 100°C. The developed length of the rod is 50 cm and is exposed to air at 30°C. The combined convective and radiative heat transfer coefficient is 30 W/m²K. Calculate :

(i) The temperature at the center of the rods

(ii) Heat transfer by the rod

(c) Derive the expression for effectiveness by NTU for Parallel Flow Heat Exchanger.

5. (a) Derive the relation for radiation b/w two infinite parallel gray bodies.

(CO6)

(b) The outer surface of a tube of 100 mm diameter is maintained at 120°C by the passage of steam through its interiors. A radiation shield is installed around the tube, with an air gap of 10 mm between the tube and the shield. The shield is at a temperature of 35°C The tube and shields are diffuse gray surfaces with emissivity of 0.80 and 0.10 respectively. What is the radiant heat transfer from the tube per unit length ? (CO6)

(c) Explain boiling and condensation with types. (CO6)

Roll No. ....

TME–502

B. TECH. (MECH.) (FIFTH SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

DESIGN OF MACHINE ELEMENTS—I

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) List the different alloys added to steel. What is the effect of adding these alloys to properties of steel ? (CO1)

   (b) How are plain carbon steels, free cutting steels, alloy steels, high alloy steels are designated as per BIS specifications ? (CO1)

   (c) Define the following mechanical properties of Engineering materials : Strength, Elasticity, Plasticity, Stiffness, Resilience, Toughness, Ductility, Malleability, Brittleness, Hardness. (CO1)

P. T. O.

2. (a) Explain the different theories of failures in brief. (CO2)

(b) It is required to design a cotter joint to connect two steel rods of equal diameter. Each rod is subjected to a tensile force of 50 kN. Design the joint and specify its main dimensions. The permissible tensile, crushing and shearing stresses are 67 MPa, 134 MPa and 33 MPa respectively. (CO2)

(c) A right angled bell-crank lever is to be designed to raise a load of 5 kN at the short arm end. The lengths of short and long arms are 100 and 450 mm respectively. The lever and the pins are made of steel 30C8 (S_{yt} = 400 N/mm²) and the factor of safety is 5. The permissible bearing pressure on the pin is 10 N/mm². The lever has rectangular cross-section and the ratio of width to thickness is 3 : 1. The length to diameter ratio of fulcrum pin is 1.25 : 1. Calculate : (CO2)

(i) The diameter and the length of fulcrum pin;

(ii) The shear stress in the pin;

(iii) The dimensions of the boss of the lever at the fulcrum; and

(iv) The dimensions of the cross-section of the lever.

Assume that the arm of bending moment on the lever extends up to the axis of the fulcrum.

3. (a) Explain the method and conditions of determining the theoretical endurance limit of a material. How is it related to actual endurance limit of the material ? (CO3)

(b) Explain the following terms : (CO3)

(i) Low and high cycle fatigue

(ii) Miner's equation of cumulative damage in fatigue

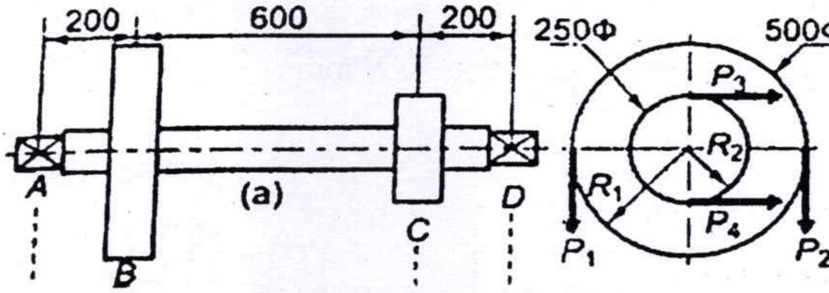
(c) Explain Goodman's, Soderberg and modified Goodman's line with neat sketch and suitable expressions. (CO3)

4. (a) (i) Explain the design criteria of shaft on the basis of strength and rigidity with numerical expressions.

(ii) Explain the design criteria of rectangular sunk key with numerical expressions

(b) The layout of a transmission shaft carrying two pulleys B and C and supported on bearings A and D is shown in Figure. Power is supplied to the shaft by means of a vertical belt on pulley B, which is then transmitted to pulley C carrying a horizontal belt. The maximum tension in belt on pulley B is 2.5 kN.

The angle of wrap for both the pulleys is 180° and the coefficient of friction 0.24. The shaft is made of plain carbon steel 30C8 (S_{yt} = 400 N/mm²) and the factor of safety is 3. Determine the shaft diameter on strength basis. (CO4)



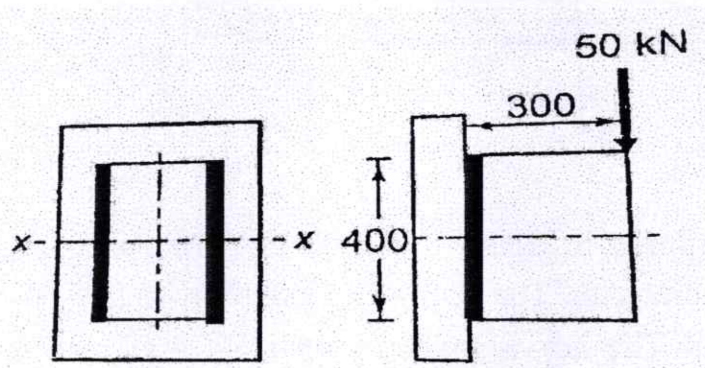
(c) It is required to design a square key for fixing a gear on a shaft of 25 mm diameter. The shaft is transmitting 15 kW power at 720 r.p.m. to the gear. The key is made of steel 50C4 (S_{yt} = 460 N/mm²) and the factor of safety is 3. For key material, the yield strength in compression can be assumed to be equal to the yield strength in tension. Determine the dimensions of the key. (CO4)

5. (a) Explain the advantages and disadvantages of riveted joint over welded joint. State the applications of riveted and welded joints. (CO5, CO6)

(b) A cylindrical pressure vessel with 1 m inner diameter is subjected to internal steam pressure of 1.5 MPa. The permissible stresses for the cylinder plate and the rivets in tension, shear, and compression are 80, 60, and 120 N/mm² respectively. The efficiency of longitudinal joint can be taken as 80% for the purpose of calculating the plate thickness. The efficiency of circumferential lap joint should be at east 62%. Design the circumferential lap joint and calculate : (CO5, CO6)

(i) thickness of the plate;
(ii) diameter of the rivets;
(iii) number of rivets;
(iv) pitch of rivets;
(v) number of rows of rivets; and
(vi) overlap of the plates.

(c) A bracket is welded to the vertical plate by means of two fillet welds as shown in Figure. Determine the size of the welds, if the permissible shear stress is limited to 70 N/mm². (CO5, CO6)



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Roll No. ....

TME–503

B. TECH. (ME) (FIFTH SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

DYNAMICS OF MACHINES

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Define and explain the superposition theorem as applicable to a system of forces acting on the system. (CO1, CO2)

   (b) Determine the required input torque on the crank of a slider-crank mechanism for the static equilibrium when the applied piston load is 1.5 kN. The lengths of the crank and the connecting rod are 40 mm and 100 mm respectively and the crank has turned through 45° from the inner-dead centre. (CO1, CO2)

   (c) Derive a relation for the turning moment at the crankshaft in terms of piston effort and the angle turned by the crank. (CO1, CO2)

P. T. O.

2. (a) What do you mean by equivalent offset inertia force? Explain in detail. (CO2, CO3)

   (b) The torque developed by an IC engine is given by

   $$T = 1000 + 300 \sin 2\theta - 500 \cos 2\theta \text{ Nm}$$

   Where is the angle turned by the crank from inner dead centre. The engine speed is 300 rpm. The mass of the flywheel is 200 Kg and radius of gyration is 400 mm.

   Determine : (CO2, CO3)

   (i) Power developed by the engine.

   (ii) Percentage fluctuation of speed with reference to the mean speed.

   (c) What is meant by cross-belt drive ? Deduce expression for the length of belt in a cross-belt drive. (CO2, CO3)

3. (a) A belt runs over a pulley of 800 mm diameter at a speed of 180 rpm. The angle of lap is 165° and the maximum tension in the belt is 2 kN. Determine the power transmitted if the coefficient of friction between the belt and pulley is 0.3. (CO3, CO4)

   (b) Deduce expressions for the friction torque considering uniform pressure and uniform wear theories for a conical collar. (CO3, CO4)

   (c) Why is meant by static and dynamic unbalance in machinery ? How can the balancing be done ? (CO3, CO4)

4. (a) A circular disc mounted on a shaft carries three attached masses of 4 kg, 3 kg and 2.5 kg at radial distances of 75 mm, 85 mm and 50 mm and at angular positions of 45°, 135° and 240° respectively. The angular positions of are mounted counterclockwise from the reference line along the x-axis. Determine the amount of the counter mass at a radial distance of 75 mm required for the static balance. (CO4, CO5)

(b) What do you mean by primary unbalance in reciprocating engines ? Explain in detail. (CO4, CO5)

(c) What are centrifugal governors ? How do they differ from inertia governors ? (CO4, CO5)

5. (a) Describe the function of a simple Watt Governor. What are its limitation ? (CO5, CO6)

(b) Explain the terms sensitiveness, hunting and stability related to governors. (CO5, CO6)

(c) What do you mean by gyroscopic couple ? Derive a relation for its magnitude. (CO5, CO6)

Roll No. ....

THV–501

B. TECH. (ME) (FIFTH SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

UNIVERSAL HUMAN VALUES : UNDERSTANDING HARMONY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) "I am the seer, doer and enjoyer. The body is my instrument." Explain. (CO2, CO3)

   (b) 'The problem today is that the desires, thoughts and expectations are largely set by pre-conditionings or sensations.' Examine this statement. (CO3, CO4)

   (c) Suggest any two programs that you can undertake to improve the health of your body. (CO4, CO5)

2. (a) 'Family is a natural laboratory to understand human relationships.' Elaborate. (CO2, CO3)

(b) Differentiate between intention and competence. How do we come to confuse between the two ? (CO3)

(c) How is trust the foundation value of relationships ? (CO3, CO4)

3. (a) Differentiate between competition and acquiring excellence with the help of one example. (CO4)

(b) What can be the basis of an undivided society–the world family ? (CO1)

(c) Right understanding in the individuals is the basis for harmony in the family, which is the building block for harmony in the society. Give your comments. (CO2, CO3)

4. (a) What do you mean by mutual fulfilment in nature ? Cite a few examples. (CO1, CO2)

(b) Suggest ways to enhance the fulfilment of human order with the other three orders. Mention any two programs you can undertake in light of the above. (CO2, CO3)

(c) What do you mean by co-existence? How are units in co-existence being in space ? (CO1, CO4)

5. (a) Where is the scope of development in nature ? How have we come to wrongly place our developmental programs ? (CO1, CO3)

(b) How does right understanding provide the basis for ethical human conduct ? Give two examples. (CO3, CO5)

(c) Choose any one dimension (education, heath, production, justice, exchange) of human endeavor in a society. Suggest what role can you play in the chosen dimension through the orientation you are going to have through your professional education. (CO2, CO3, CO5)

Roll No. ....

DETF–02

B. TECH. (ME) (FIFTH SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

POWER PLANT ENGINEERING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Define the following in brief and also write their physical significance in power plant engineering : (CO1, CO4, CO5)

(i) Load Factor,

(ii) Diversity factor,

(iii) Load curve,

(iv) Load duration Curve, and

(v) Capacity factor.

(b) A power station has to supply load as follows : (CO1, CO4, CO5)

Time (hours)	Load (MW)
0—6	45
6—12	135
12—14	90
14—18	150
18—24	75

(i) Draw the load curve.

(ii) Draw load duration curve.

(iii) Choose suitable generating units to supply the load.

(iv) Calculate the load factor.

(v) Calculate the plant capacity factor.

(c) The maximum demand of a power station is 96000 kW and daily load curve is described as follows : (CO1, CO4, CO5)

Time (Hours)	Load MW
0—6	48
6—8	60
8—12	72
12—14	60
14—18	84
18—22	96
22—24	48

(i) Determine the load factor of power station.

(ii) What is the load factor of standby equipment rated at 30 MW that takes up all load in excess of 72 MW ? Also calculate its use factor.

2. (a) Explain with the help of a neat diagram the arrangement and working of Fluidized Bed combustion (FBC) system. Also explain its types with neat sketches. (CO2, CO3, CO4)

   (b) A chimney of height 32 m is used for producing a draught of 16 mm of water. The temperatures of ambient air and flue gases are 27°C and 300°C respectively. The coal burned in the combustion chamber contains 81% carbon, 5% moisture and remaining ash. Neglecting losses assuming the value of burnt products equivalent to the volume of air supplied and complete combustion of flue find the percentage of excess air supplied. (CO2, CO3, CO4)

   (c) Discuss the differences in sub critical, supercritical and ultra supercritical boilers. Also, explain the natural circulation and once through system with neat sketches only. (CO2, CO3, CO4)

3. (a) What is a nuclear reactor? Describe the various parts of a nuclear reactor with neat sketches and by mentioning their role in brief. (CO2, CO5, CO1)

   (b) Describe Boiling water reactor (B.W.R.) and Pressurised water reactor (P.W.R.) with the help of neat sketches. (CO2, CO5, CO1)

   (c) 200 MW of electrical power (average) is required for a city. If this is to be supplied by a nuclear reactor of efficiency 20 per cent, using U^{235} as the nuclear fuel, calculate the amount of fuel required for one day's operation. Assume that energy released per fission of U^{235} nuclide = 200MeV. (CO2, CO5, CO1)

4. (a) What is hydro-power? What are the advantages and disadvantages of a hydro-electric power plant. Also, write its detailed classification (names only). (CO2, CO3, CO4)

(b) The quantity of water available for hydroelectric station is 250 m^3/ sec under a head of 1.6 m. If the speed of the turbine is 50 r. p. m. and efficiency 82% determine the number of turbine units required. Take specific speed as 740. (CO2, CO3, CO4)

(c) Explain Pelton wheel turbine. Derive expressions for its hydraulic, mechanical, and overall efficiency. (CO2, CO3, CO4)

5. (a) Draw a general layout of a gas turbine plant and name the major components. Also, discuss its advantages and disadvantages. (CO1, CO2, CO6)

(b) Draw a general layout of a diesel power plant and name the major components. (CO1, CO2, CO6)

(c) The brake thermal efficiency of a diesel engine is 30 per cent. If the air to fuel ratio by weight is 20 and the calorific value of the fuel used is 41800 kJ/kg, what brake mean effective pressure may be expected at S. T. P. conditions ? (CO1, CO2, CO6)

Roll No. ....

DEEM–03

B. TECH. (FIFTH SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

MANAGEMENT OF INTELLECTUAL PROPERTY RIGHTS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Develop and comprehensive strategy for a startup to maximize the value of its patent portfolio. (CO3 & CO4)

   (b) How does the 'non-obviousness' requirement in patent law affect the patentability of inventions and why is it a crucial elements in patent examination ? (CO3 & CO4)

   (c) How do patents contribute to fostering innovation in various industries ? (CO3 & CO4)

P. T. O.

2. (a) Define IP audit. Also, mention the different types of valuation methods. (CO2 & CO5)

   (b) How do industrial designs contribute to product differentiation in the market ? (CO2 & CO5)

   (c) Explain the role of industrial designs in the automobile sector. (CO2 & CO5)

3. (a) What are the basic elements of patentability ? Discuss. (CO5 & CO6)

   (b) Define 'inventive step' under the Indian Patent Act. List down the various types of patents. (CO5 & CO6)

   (c) Illustrate the process of creating and registering and industrial design for a new product. (CO5 & CO6)

4. (a) What constitutes a trademark and what types of marks can be registered ? (CO4 & CO5)

   (b) Cay you analyze the impact of trademarks on brand recognition and consumer trust ? (CO4 & CO5)

   (c) How do geographical indications differ from trademarks and patents ? (CO4 & CO5)

5. (a) What does copyright protect and what are the typical durations of copyright protection ? (CO1 & CO2)

   (b) What are the key components of an effective intellectual property management strategy ? (CO1 & CO2)

   (c) How do different types of intellectual property rights (patents, trademarks, copyright, etc.) require distinct management approaches ? (CO1 & CO2)

Roll No. ....

DEDE–04

B. TECH. (ME) (FIFTH SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

PYTHON FOR DATA SCIENCE

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Discuss the concept of functions in Python. Explain the difference between built-in functions and user-defined functions. Provide an example of a user-defined function, including its definition and usage. (CO1)

(b) Write a Python program to find the most frequent word in a string. (CO1)

(c) Write a Python function to calculate the fuel efficiency of a car in kilometers per liter (km/L) . The function should take the distance traveled in kilometers and the amount of fuel consumed in liters as input and return the fuel efficiency in km/L. (CO1)

2. (a) Define and explain the concept of lambda expressions (anonymous functions) in Python. Discuss situations where lambda expressions are useful, and provide an example of a lambda function that performs a specific task. (CO2)

   (b) Write a Python function to reverse the order of a dictionary (swap the keys and values). (CO2)

   (c) Create a Python function that takes two tuples representing coordinates (x, y) as input and calculates the distance between the two points. (CO2)

3. (a) Discuss the role of the NumPy library in performing linear algebra operations. Explain the importance of NumPy functions for matrix multiplication. (CO3)

   (b) Write a Python program that creates a 1D NumPy array and reshapes it into a 2D array of dimensions (3, 4). Then, transpose the 2D array and print the result. (CO3)

   (c) Write a Python program that creates a 4×4 NumPy array and multiplies it by a 1D NumPy array of size 4. Ensure broadcasting is used correctly, and print the resulting array. (CO3)

4. (a) What is the role of Index objects in Pandas ? How do Index objects facilitate data manipulation and selection in Series and DataFrames ? Provide an example to demonstrate their importance. (CO4)

(b) Data on the mechanical properties of various materials used in mechanical engineering applications is as follows : (CO4)

Material	Young's_Modulus (GPa)	Tensile_Strength (MPa)
Steel	210	500
Aluminium	70	300
Copper	120	250
Titanium	110	600
Brass	90	100

Set the 'Material' column as the index of the DataFrame, Calculate and display the mean, median, and standard deviation of Young's Modulus and Tensile Strength.

(c)

Aircraft_Type	Maximum_Speed (m/s)	Maximum_Altitude (m)
Boeing 747	294.5	13716
Airbus A380	291.9	13106
F-22 Raptor	771.7	19812
Cessna 172	62.6	4267
Gulfstream G650	316.2	15545

Generate the code for the following : (CO4)

(i) Set the 'Aircraft_Type' column as the index of the DataFrame.

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(ii) Calculate the mean, median, and standard deviation of the maximum speeds and maximum altitudes for the aircraft models.

(iii) Identify the aircraft with the highest maximum speed and the one with the highest maximum altitude.

5. (a) What is the purpose of feature scaling in machine learning, and what are common techniques for feature scaling ? (CO4)

(b) Create a 2 × 2 grid of subplots using Matplotlib. In each subplot, plot a sine wave with a different color and linestyle. Provide a legend indicating the color and linestyle for each subplot. (CO5)

(c) Create a distribution plot using Seaborn for a given dataset's numerical feature, and include a KDE plot for visualizing the probability density. You may choose any dataset. (CO5)

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